INTERVIEW-PREP PACKET 2 OF 5 · PER-INTERVIEWER · BUSINESS-PROCESS LENS

Amazon NACF Supply Chain loop — Packet 2: Ryan Hoerger

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SHADOW INTERVIEWER: SANDY HSIAO (A SECOND NOTE-TAKER — SAME PACE, SAME NUMBERS; EXPECT ONE QUESTION FROM THE SHADOW NEAR THE END)

> **Strategy in one line.** The most quantitative interviewer on the panel. Tell two Plan-process redesigns with the math visible and every input labeled: S6 (honest supply definition → shortage alert a lead-time early → $2M released under a two-signal rule) and S8 (brochure lead times → measured lane distributions → p99 promise → ~$1.5M released with zero stockouts). The two programs' dollars never meet.

LP focus — my guess, not confirmed: Dive Deep · Are Right, A Lot · Insist on the Highest Standards or Learn & Be Curious. Amazon does not disclose the assignment; the guesses are reasoned from the interviewer's title and the JD in "Read me first" and should be held loosely. No interviewer biography is asserted anywhere in this packet — only title-based inference, labeled as such.

Shadow etiquette: address both; the shadow is usually a new interviewer being calibrated — same numbers, same pace; if the shadow asks the last question, answer it with the same discipline (phase asked, then stop).

Canon: amazon/amazon-l6-scm-master-packet.html (§1.2 roster, §4.1 ledger). Every number below is that packet's value; every dollar carries its basis label (record · user anchor · modeled · priced).

Read me first

The interviewer's lens (inference from title + JD — not a biography)

Inference from the title. "Placement Analytics" is the analytical arm of inventory placement — deciding where inventory should sit in the network so that the promise holds at the lowest landed cost. A manager in that function lives on forecast error, lead-time variability, safety-stock policy, service levels and the data-quality underneath them. He will be the most quantitative interviewer on the panel and the most likely to ask "how was that measured?", "what was the denominator?", and "what is the softest input in your model?" He is also the one most likely to recognise a textbook formula being passed off as a program artifact — so the honesty labels matter more here than anywhere.

What that means for the answers. Two Elementum planning stories, told as Plan-process redesigns with the math visible: the shortage-alert / real-time supply-demand view (S6 — forecast accuracy 70 → 85%, $2M safety stock released under a two-signal rule) and the measured-lane lead-time program (S8 — promise at p99 on 200 lanes, 580K units / ~$1.5M released with zero stockouts). Both end on *capital released on evidence with service held* — asset utilization and reliability in the JD's words. Keep the two programs' dollars apart in every sentence. A shadow (Sandy Hsiao) is on the call: same pace, same numbers, expect one question from the shadow near the end.

Three guessed LP focus areas, with rationale

> **Labeled guess.** Amazon assigns 2–3 Leadership Principles per interviewer and does not disclose them. These three are inferred from the title and the JD; if the first question lands on a different LP, the twist maps under each story carry the pivot.

GUESSED LP	WHY THIS INTERVIEWER WOULD CARRY IT	STORY TO OPEN · BACKUP
Dive Deep	An analytics manager tests whether the candidate goes to the data grain — measured lane distributions instead of brochure values; the definition of "supply" before any screen. S8 is the Dive Deep primary in the master map.	S8 (open) · S6 (the supply definition, the MAPE/bias measurement)
Are Right, A Lot	Placement decisions are judgment under uncertainty; he will probe how the candidate was right against the room's belief and how the decision was made safe (release rule, watch window, fat-tail exception).	S8 (measured vs brochure; p99 chosen by planning on the asymmetry) · backup: S6 (refusing unfirmed MRP orders as supply)
Insist on the Highest Standards / Learn & Be Curious (one of the two)	The 99% lead-time-accuracy bar before any release; the two-signal rule before any buffer came out; the MIT toolkit acquired to design what I had been implementing — labeled as toolkit, never as day job.	S8 (99% bar) · S6 (both-errors-improved rule)

Answer strategy — business first, systems as enablers

- **Open with the process problem and its cost, not the system.** First sentence names the broken process (Plan, Source, Make, Deliver or Return), who it hurt (customer, Finance, the floor) and the baseline number. The platform appears only as the enabler of the redesign, in one clause.
- **Use the JD's four outcome words as the results frame:** reliability (did the promise hold?), responsiveness (how much earlier did we see it?), cost efficiency ($ with its basis label), asset utilization (capital, labor-hours, capacity released).
- **Action = operating-model redesign in gated phases** — Discovery → Plan & Design → Build & Test → Cutover → Hypercare, each spoken as entry bar · what I delivered · exit bar. Say the gate that fired, not the

meeting that happened.

- **Every dollar carries its basis word in the same breath** — record · user anchor · modeled · priced. A record dollar is said flat; a modeled dollar is said with its inputs. Never promote a modeled number to record.
- **Program-lead voice.** I owned definitions, rules, gates and the signature chain; engineers, SIs and vendor teams configured and built to the spec I signed. The only platform personally configured in canon is Manhattan DOM.
- **The pitch ends on the primary metric.** Then stop. Silence after the pitch is the interviewer filling their STAR worksheet; their probes are the second half of the answer.
- **For Placement Analytics specifically:** lead with the planning parameter that was wrong (brochure lead time; unfirmed orders counted as supply), the decision it corrupted (buffer size, shortage response), and the rule that released capital safely. Show one formula only when asked, labeled as the MIT toolkit illustration — the data team built the distributions; planners sized the buffers; Finance signed.
- **Keep S6 and S8 apart:** $2M (S6, forecast-driven, two-signal rule) and ~$1.5M / 580K units (S8, lead-time-driven, 99% bar) are different programs — never add them, never swap their mechanisms. "Six sites" — never "six NA sites"; never paste S6's "3 months" onto the six-site ingest (that is S7, not in this packet).

> **Loop-consistency rule.** One question, one story; never the same story to two interviewers — interviewers compare notes. Held for other packets (do not spend here): S11/S12 (Tripathy) · S2/S18 (Goforth) · S1/S3 (Lan) · S16/S15/S10/S14 (Josyula). S7 (CM BOM) and S9 (EDI Pareto) are unallocated master-packet backups — pointers only, and no BOM-explosion content.. If pushed for "another example" on an LP, use the named backup in the question bank, never a story allocated to another interviewer.

The stories — business-first, canon numbers

Two Elementum planning stories from the same six-site manufacturer. Each: 60-second pitch → Situation/Task → Action in gated phases → Results → Probing tuned to a placement-analytics manager → the key math → Twist map.

S6 — Elementum — one real-time supply/demand view across eight data domains: a shortage visible a lead-time early, forecast accuracy 70% → 85%, $2M of safety stock released under a two-signal rule

LPs this story can serve here: Dive Deep (the definition of supply; MAPE/bias at the decision grain) · Earn Trust (the two-signal release rule signed with Planning and Finance) · Deliver Results (third — alert live in 3 months) · Think Big (one view over eight domains) · Are Right, A Lot (backup — refusing unfirmed orders as supply)
Process family: Plan — S&OP / shortage management / safety-stock policy; data-integrity program behind the plan

Honesty label. I owned the definitions, the data-integrity program and the delivery; the client's SAP teams and our integration engineers built the feeds; planners owned the forecast — I owned its measurement and the release rule. I have not owned a demand forecast end-to-end and won't claim it. The safety-stock formula is how I explain it now with the MIT toolkit — an illustration, not a program artifact.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. A six-site manufacturer — mixed make-to-order and make-to-stock — learned about a component shortage when the line stopped. S&OP ran off private spreadsheets, shipment data was 60% complete, and supplier, material, BOM, inventory, production, transportation, sales-order and planning data lived in SAP and site systems but never met. Leadership's question — which customer orders are at risk this quarter? — had no data path, so the plan responded with expedites instead of re-plans and buffers sized on noise. **Task.** Nobody owned the seam; I took the definitions, the data-integrity program and the delivery. **Action.** I defined supply honestly before any screen — on-hand plus production plan plus released planned orders only, at SKU × location × date — and refused to count unfirmed MRP orders because they change every run. Domains went live in waves sequenced by planner trust, each behind a published data-quality gate, with daily reconciliation to the SAP source and false alarms worked as defects. The shortage alert fires when days-of-cover falls to the lane's measured p90 lead time — the last moment a re-plan is cheaper than a flight. Safety stock came out only under a rule Planning and Finance signed: release where forecast error and lead-time error had both improved. **Result.** Primary metric: forecast accuracy on the scoped class-A SKUs 70% → 85%. $2M of safety stock released under the two-signal rule — record, one-time working capital — with ≈$0.8M a year of modeled recurring value around it; the shortage alert live in three months; planner effort −50%. That $2M is a different program from the lane program's $1.5M; I never add them.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — A strategic manufacturing customer with six sites: planners learned about a component shortage when the line stopped; S&OP ran off private spreadsheets; shipment data was 60% complete; eight data domains lived in SAP and site systems but never met. Leadership's question — "which customer orders are at risk this quarter?" — had no data path.

The Plan process underneath was running on an unsafe definition of supply (unfirmed MRP orders counted because they looked better), an unmeasured lead-time error, and a shortage signal that arrived at line-down. The cost: expedites instead of re-plans, buffers sized on noise, and planners spending ~40% of their week reconciling rather than root-causing.

T — Nobody owned the seam — no spec, no agreed grain, no definition of "supply," no data-quality bar. I chose all four, as owner of the definitions, the data-integrity program and the delivery. Goal: a real-time supply/demand view with honest definitions and a shortage visible a lead-time early — and planner trust, which was the real product.

Baselines on the table: forecast accuracy 70% on scoped class-A SKUs · shortage detected at line-down (target: a lead-time early) · shipment-data completeness 60% · planner effort on reconciliation and firefighting (≈40% of the week, illustrative share; the −50% is record).

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (4–6 wks)** — profiled every domain (completeness, latency, joinability) to my own gap templates; wrote the definitions memo — grain SKU × location × date; supply = on-hand + production plan + released planned orders only — and the metrics tree (OTIF, inventory $, MAPE/bias, lane lead-time accuracy, completeness, latency, adoption). Refused Planning's request to count unfirmed MRP orders as supply. *Exit bar:* signed gap findings per domain; scope frozen in writing; planners agree the supply definition before any screen.
- **Plan & Design (parallel)** — master-data foundation plan; MPN mapping worklist with its coverage gate; domain-wave sequence by trust; DQ scorecard spec; the 10-step onboarding playbook. *Exit bar:* master-data gates green; unmapped-MPN list under threshold with a named owner; wave order signed by domain owners — who could fail their own wave.
- **Build & Test (8–12 wks)** — domain waves behind published DQ gates — inventory and production first (numbers planners already believed), then purchasing/ASN, then transportation; shortage-alert rule specified (days-of-cover vs lane lead-time p90); daily reconciliation to MD04/COOIS/VA05/MB52; weekly root-cause loop with false alarms worked as defects. *Exit bar:* per-domain completeness ≥ gate; reconciliation passing daily; false alarms trending down; pilot planners using the alert.
- **Cutover & Go-Live (4 wks)** — views live for pilot planners with the at-risk list replacing the Monday spreadsheet; nobody mandated off the spreadsheet — rollback stayed free. *Exit bar:* planners answer "where is my product" from the platform in the live S&OP meeting.
- **Hypercare / Close (4 wks)** — exception workbench staffed; weekly root-cause reviews handed to domain owners; the safety-stock release rule signed with Planning and Finance — release only where forecast error AND lead-time error both improved. *Exit bar:* blockers/day under threshold two consecutive weeks; the release survives audit.

Influence and ownership. Mine: the definitions memo, the metrics tree, the DQ scorecard spec, the wave sequence, the release rule and the onboarding playbook. The client's SAP teams and our integration engineers built the feeds; planners owned the forecast — I owned its measurement (MAPE/bias per SKU family) and the release rule; domain owners held sign-off over their own gates. The MPN worklist gated longer than planned — escalated to the sponsor as a re-sequenced date with the coverage number attached, not a slip.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — $2M safety stock released (record, one-time working capital, under the both-errors-improved rule signed with Planning and Finance) + ≈$0.8M/yr *modeled* recurring (carry ≈$0.2M at 10% · expedites pre-empted ≈$0.3M · planner capacity ≈$0.3M; range $0.5–1.2M) — never added to S8's $1.5M.
- **#** — eight data domains into one view across six sites; shortage-alert product live in 3 months; adoption 8× (record).
- **%** — forecast accuracy 70% → 85% on scoped class-A SKUs (record); planner effort −50% (record); shipment data 60% complete at the start.
- **Time** — the alert fires at days-of-cover ≤ lane lead-time p90 — a lead-time early instead of at line-down (record mechanism); views live in 3 months (the alert product; the six-site ingest is a separate 5-month program — never paste 3 months onto it).
- **Customer** — "which customer orders are at risk this quarter?" answered from the platform in the live S&OP meeting, not a spreadsheet (record exit bar).
- **Quality** — daily reconciliation to MD04/COOIS/VA05/MB52; false alarms classified weekly to root cause and burned down as defects (record).

PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

Analytics — How was 70 → 85% measured, and on what?

Accuracy = 1 − MAPE on the scoped class-A SKUs, with bias published alongside it from month two, at the SKU-family grain the planners decide on. The planners owned the forecast; I owned the measurement and the input integrity — I have not owned a demand forecast end-to-end and won't claim it.

Analytics — Why exclude unfirmed MRP planned orders from supply? That understates it.

Deliberately. Unfirmed orders breathe every MRP run; counting them amplifies noise and inflates supply, which is exactly how the line-down surprises happened. Released planned orders only — accepted by planners before the first screen was built.

Analytics — The $2M — how do I know it is not a prettier average?

Because of the release rule: buffer came out only on SKU families where forecast error and lead-time error had both demonstrably improved, signed by Planning and Finance. That is why the release survived audit. It is one-time working capital; the recurring value around it is my model and labeled.

Analytics — Your expedite row is the softest. Defend it.

Agreed — about sixty expedites a year pre-empted at a $5K premium, both illustrative. The record is the mechanism: the alert fires at days-of-cover ≤ lane p90. To firm it I would pull the expedited-freight ledger for the six sites twelve months before and after the alert went live and count the premium directly.

Analytics — Why p90 for the alert threshold and p99 for the promise (S8)?

Different decisions. The alert is a re-plan trigger — p90 gives the planner a window before the tail bites. The promise is a commitment — it sits at the tail you can defend. Same measured distribution, two policies read off it.

Shadow / closing — What would you do differently?

Start the MPN-mapping worklist on day one — it is never "done" and it gated more than I planned. And define supply honestly before any screen, which held.

THE KEY MATH — THE NETTING WALK AT FINISHED-GOODS GRAIN, THE ALERT RULE AND THE TWO-SIGNAL RELEASE (ILLUSTRATIVE VALUES; RECORD PERCENTAGES)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Accuracy before / after	`1 − 0.30 = 70% → 1 − 0.15 = 85%`	record	70% → 85%
2	Honest supply	`supply(t) = on-hand + in-transit ≤ t + production ≤ t + RELEASED planned orders`	record rule	unfirmed MRP orders excluded
3	Projected balance (FG SKU)	`PAB(3) = 400 − 500 = −100; PAB(7) = 1,000 − 1,200 = −200; PAB(10) = 1,800 − 1,600 = +200`	illustrative FG example — timing shortage, not volume	fix window = day 9 − day 7 = 2 days
4	Alert rule	`DoC = 1,200 / 100 = 12 days ≤ lane p90 14 → alert; p90 = 9 → no alert`	record design; values illustrative	a lead-time early, no false alarm on the fast lane
5	Two-signal release	`SS = z·d·σ_L = 2.33 × 100 × 4 = 932 → σ_L 2: 466; release 466 units — only if BOTH errors improved`	toolkit illustration; the rule is record	release survives audit
6	Recurring value	`$2M × 10% = $0.2M · 60 × $5K = $0.3M · 2.4 FTE × $120K = $0.3M`	modeled (benchmarks labeled)	≈$0.8M/yr; $2M one-time record

MRP math at finished-goods grain only — no BOM explosion in this packet. The six-site ingest (S7) is a separate program; never paste this story's "3 months" onto it.

Where this lands at Amazon. Placement rests on two error signals — forecast error and lead-time error — measured at the decision grain. The transferable disciplines: define supply before the screen, alert at the last moment a re-plan is cheaper than an expedite, and release buffer only when both error signals improve. Reliability (shortage seen a lead-time early), cost efficiency (expedites pre-empted), asset utilization ($2M of working capital released on evidence).

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S6

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you had to dig into the data to find the real problem.	Dive Deep	The supply definition — unfirmed orders inflating supply — found by profiling eight domains; MAPE and bias at the grain planners decide on.	Skip the SAP table names unless asked; systems are enablers.
Tell me about a time you released inventory / improved working capital.	Frugality / Deliver Results	$2M under the two-signal rule — record, one-time; the recurring ≈$0.8M labeled modeled.	Never add the $2M to S8's $1.5M; never claim $2M per year.
Tell me about a time you were right when the business disagreed.	Are Right, A Lot	Refusing unfirmed MRP orders as supply because it looked better; refusing a blanket safety-stock release — agreed as a rule instead.	Do not make Planning the villain; the rule did the refusing.
How do you get planners to trust a new signal?	Earn Trust	Waves sequenced by trust (inventory and production first); false alarms worked as defects; nobody mandated off the spreadsheet; domain owners could fail their own wave.	Do not claim adoption dollars; adoption 8× is the record.
How would you design a shortage / at-risk-order signal for the network?	Functional — Plan process design	Grain, honest supply, days-of-cover vs measured lane p90, DQ gate per feed, two-signal release — the §3.99 opening: book of record and metric first.	No BOM explosion; finished-goods grain only.
What's the softest number in your model?	Highest Standards / Earn Trust	The expedite row (60 × $5K); how I would firm it from the expedited-freight ledger.	Never present planner capacity as cash — it is time redeployed.

S8 — Elementum — replacing brochure lead times with measured lane distributions: promise at p99 on 200 lanes, 99% lead-time accuracy, 580K units / ~$1.5M of safety stock released with zero stockouts

LPs this story can serve here: Dive Deep (primary — measured the distribution, not the average) · Are Right, A Lot (backup — measured vs brochure; p99 on the asymmetry) · Frugality (the release) · Insist on the Highest Standards (the 99% bar before any cut) · Earn Trust (Finance signed each lane; the exception written into the rule)

Process family: Plan / Source / Deliver seam — inbound lead-time policy, safety-stock sizing, promise policy; parameter governance with a drift flag

Honesty label. I owned the requirement (measured lanes, not brochures), the p99 promise policy and the release rule; the data team built the distributions; planners sized the buffers; Finance signed the release. Lane-cohort counts and week markers are not in the record. The safety-stock formula is the MIT inventory-theory toolkit as I explain it now — an illustration, not a program artifact. Depends on the six-site canonical model (S7 — pointer only).

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. The same six-site manufacturer planned inventory on carrier-brochure lead times — every buffer sized on a variability nobody had measured: too much on reliable lanes, too little on fat-tailed ones. The logistics manager asked for a "lead-time variability report." **Task.** I reframed it as a parameter and policy problem — nobody owned the seam between transportation reality and planning parameters, and I took it — scoped at 200 lanes by site, lane and mode. **Action.** I wrote the requirement to the data team: mean, spread and tail per lane from actual production cycle plus inbound transit; defined the promise at p99 cumulatively — the smallest day d with $P(L \le d) \ge 0.99$, not average-plus-padding — and showed each lane's asymmetry, line-down cost against carrying cost, and let planning choose the percentile; they chose p99. The release rule was written as a signed planning act with the exception inside it: fat-tail lanes keep their buffer until the source process moves; Finance agreed the rule in the same room, before any cut. I would not endorse a release until lead-time accuracy hit 99% on the 200-lane set; the stockout-and-expedite watch was armed first; every parameter got a drift flag and a refresh cadence so the promise never decays into a new brochure. **Result.** Primary metric: lead-time accuracy on 200 lanes — brochure values → 99% at the p99 promise. 580K units, about $1.5M of safety stock, released without a stockout — record, one-time working capital — with ≈$0.6M a year modeled around it. A different program from the $2M forecast release; never added.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — The six-site manufacturer's logistics manager asked for a "lead-time variability report." The real problem: inventory planning ran on carrier-brochure lead times — every buffer sized on a variability nobody had measured; too much buffer on reliable lanes, too little on fat-tailed ones; the promise to the plan was folklore.

The risk cut both ways: keep planning on brochures and you hold too much on reliable lanes and stock out on fat-tailed ones; cut buffer without evidence and the stockout is yours. And an unmaintained promise decays into a new brochure — which is why every parameter needed a drift flag.

T — Nobody owned the seam between transportation reality and planning parameters; I took it. Goal: measured lanes replace catalog values; a promise at a defensible tail; buffer released only on evidence, with service held — scoped at 200 lanes by site, lane and mode.

Baselines on the table: lead-time parameters = carrier-brochure values, variability unmeasured · buffer capital sized on catalog values · no lead-time-accuracy measurement · target 99% lead-time accuracy on the 200-lane set before any release.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — reframed the report request as a parameter problem; scoped 200 lanes by site/lane/mode; wrote the success criteria — measured lanes replace catalog values; buffer released only on evidence; fat tails keep buffer; showed one right-skewed lane where the average hid the tail. *Exit bar:* planning and logistics agree the problem is the parameter, not the report.
- **Plan & Design** — requirement to the data team (mean, spread, tail per lane; production cycle + inbound transit vs actual); the p99 promise policy (cumulative, not μ + 2.33σ); the release rule as a signed planning act with the fat-tail exception inside it; the stockout-and-expedite watch; drift flag and refresh cadence. On p95 vs p99, showed the asymmetry per lane and left the choice to planning. *Exit bar:* policy signed by planning; Finance agrees the release rule in the room; data team accepts the spec.
- **Build & Test** — data team built the lane distributions and tracked lead-time accuracy per lane; manual reconciliation stayed alive. *Exit bar:* lead-time accuracy 99% on the 200-lane set — the bar I would not endorse a cut below.
- **Cutover & Go-Live** — planners executed the buffer release lane by lane under the signed rule; fat-tail lanes excluded until their source process moved; watch armed first. *Exit bar:* 580K units / ~$1.5M released; no stockout in the watch window.
- **Hypercare / Close** — read the watch; set drift flags per lane; handed the refresh cadence and the release rule to planning as an act they could re-apply. *Exit bar:* watch clean; the promise stays measured.

Influence and ownership. I owned the requirement, the promise policy and the release rule; the data team built the distributions and tracked accuracy; planners sized the buffers and executed; Finance signed each lane. When Finance pushed for a broader release, the rule they had signed did the refusing — fat-tail lanes keep buffer until the source process moves. The percentile fight never happened: I showed each lane's line-down cost against carrying cost and planning chose p99 — their choice surviving an audit is worth more than my choice winning a meeting.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — 580K units / ~$1.5M of safety stock released without a stockout (record, one-time working capital); ≈$0.6M/yr *modeled* recurring around it (carry ≈$0.3M at a benchmark 20% all-in rate + expedites avoided ≈$0.3M; range $0.3–1.4M) — a different program from S6's $2M; never added.
- **#** — 200 lanes measured by site, lane and mode; 580K units released (record); ≈$2.59/unit derived — low-value components.
- **%** — lead-time accuracy 99% on the 200-lane set at the p99 promise — the bar below which I would not endorse a cut (record).
- **Time** — release only after the 99% bar; watch window clean; refresh cadence and drift flag per lane handed to planning (record).
- **Customer** — service held — zero stockouts on released lanes through the watch window (record).
- **Quality** — release only under the signed rule; fat-tail lanes kept buffer until the source process moved; the promise stays measured, never a new brochure (record).

PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

> **Analytics — Why p99 and not p95? That is a lot of buffer.**
>
> Per lane, the cost of a line-down on a fat-tailed lane dwarfed the carrying cost of the extra day. I showed the asymmetry lane by lane and let planning choose — they chose p99. On reliable lanes the measured σ was so far below the brochure that p99 still released buffer; on fat-tailed lanes the buffer stayed.

> **Analytics — How did you compute p99 on skewed lanes?**
>
> Cumulatively from the empirical distribution — the smallest d with $P(L \le d) \ge 0.99$ — not μ + 2.33σ. On one right-skewed lane the normal shortcut understated the true p99 by about 3.4 days; that lane is why the definition is cumulative.

Analytics — Is the $1.5M measured?

The release is record: 580K units out, zero stockouts in the watch window, Finance signed each lane. The ≈$2.59/unit is the derived average. What is modeled is the recurring value around it — carry at 20% all-in (capital-only would be ≈$150K at 10%) and avoided expedites — each input labeled.

Analytics — The expedite number assumes what?

Ten thousand shipments a year on the 200 lanes and one late arrival in ten forcing an expedite before p99 — illustrative. I would pull the expedite ledger by lane for the year before and after and compare directly.

Analytics — What is lead-time accuracy, exactly?

1 − |actual − promised| / promised, tracked per lane against the p99 promise; the bar was 99% on the whole 200-lane set. Below it, no release — written into the act before anyone asked.

Shadow / closing — How does the promise stay honest after you leave?

Drift flag per lane with a refresh cadence: p99 is recomputed on refresh, and the buffer decision re-opens if accuracy falls. The stockout-and-expedite watch became a standing report owned by planning.

THE KEY MATH — ONE RELIABLE LANE, ONE FAT-TAIL LANE, AND THE RECURRING VALUE (ILLUSTRATIVE UNITS; RECORD PERCENTAGES AND THE RECORD RELEASE PAIR)

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Brochure-sized buffer	`SS = z·d·σ_L = 2.33 × 500 × 5 = 5,825 units`	illustrative; toolkit form	5,825 units of folklore
2	p99 defined cumulatively	`p99 = min{d : P(L ≤ d) ≥ 0.99}`	record definition	normal shortcut understates a skewed lane by 3.4 days
3	Reliable lane re-sized	`2.33 × 500 × 1.2 = 1,398; release = 5,825 − 1,398 = 4,427`	measured σ = 1.2 vs brochure 5 (illustrative)	4,427 units off this lane
4	Fat-tail lane	`2.33 × 500 × 6.5 = 7,573 — buffer KEPT`	measured σ = 6.5 > brochure 5 (illustrative)	kept until the source process moves
5	Accuracy gate	`LTacc = 1 − |actual − promised| / promised; bar 99% on 200 lanes`	record	99% held before any cut
6	The release	`$1,500,000 / 580,000 = $2.59/unit`	record pair; average derived	580K units / ~$1.5M, 0 stockouts
7	Recurring value	`$1.5M × 20% = $0.3M; 900 fewer late × 10% × $3K ≈ $0.27M`	modeled (benchmarks labeled)	≈$0.6M/yr; never added to S6's $2M

Depends on the six-site canonical model (S7) — six definitions cannot produce one σ. S7 is a pointer only in this packet; say "six sites," never "six NA sites."

Where this lands at Amazon. Inbound lead-time variability is half of every buffer and every promise in a placement model. Transferable: measure the lane's own distribution, promise at the tail you can defend, release capital only on evidence with the exception written into the rule, and give every parameter a drift flag. Reliability (zero stockouts), cost efficiency (expedites avoided), asset utilization (~$1.5M of working capital out).

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S8

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time you went deeper than the request.	Dive Deep	A "variability report" reframed as a promise-and-buffer policy; mean, spread and tail per lane from actuals; p99 defined cumulatively.	Do not claim to have built the distributions — the data team did, to my requirement.

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about a time the data contradicted conventional wisdom.	Are Right, A Lot	Brochure values vs measured lanes; one right-skewed lane where the average hid the tail; planning chose p99 on the asymmetry.	S10 (Austria — "the noise said carriers") is the bar raiser's; do not use it here.
Tell me about a time you held a standard under pressure.	Insist on the Highest Standards	No release below 99% lead-time accuracy; Finance's push for a broader release answered with the rule they signed.	Do not frame Finance as the obstacle; the exception was in their rule.
How do you set safety stock / service levels?	Functional — placement policy	Two error signals (forecast, lead time), measured at the lane × SKU grain; policy by cost asymmetry; release by rule with a watch window; drift flags.	Never present the formula as a program artifact — MIT toolkit illustration.
What did you refuse to do?	Have Backbone	Averaging away a tail; endorsing a cut below 99%; a blanket release — all written into the act before anyone asked.	Keep the $2M (S6) out of this answer.
How did you make Finance comfortable releasing capital?	Earn Trust	Finance agreed the release rule in the same room as planning signed the policy — before any cut; signed each lane; the watch armed first.	No annualized $1.5M — it is a one-time release.

Question bank — likely questions for this interviewer, with a one-line answer plan

Behavioral and functional questions a Placement Analytics manager is likely to ask. Each plan points at a packet story or a named backup in the master packet; stories allocated to other interviewers are pointers only.

#	LIKELY QUESTION	ANSWER PLAN (ONE LINE)	STORY
1	**Walk me through how you improved forecast accuracy.**	S6 pitch — honest supply definition first, MAPE/bias at the decision grain, waves by trust; 70 → 85% on scoped class-A SKUs; planners owned the forecast, I owned its measurement.	S6
2	**Tell me about a time you used data to change an inventory policy.**	S8 pitch — brochure → measured lanes, p99 promise, 99% bar, 580K units / ~$1.5M out with zero stockouts.	S8
3	**How do you decide where / how much safety stock to hold?**	Two error signals measured at the grain; cost asymmetry sets the percentile; release only by signed rule with a watch; fat tails keep buffer.	S8 → S6
4	**What is the softest assumption in your biggest result?**	S6: the expedite count (60 × $5K). S8: the late-arrival share. Say how each would be firmed from a ledger.	S6 / S8
5	**How do you measure lead-time performance?**	Lead-time accuracy = 1 − |actual − promised| / promised against a per-lane p99 promise; drift flag and refresh cadence.	S8
6	**Tell me about a time a stakeholder wanted a number that looked better.**	Planning's request to count unfirmed MRP orders as supply — refused; released planned orders only.	S6
7	**How did you handle a dependency that slipped (MPN mapping)?**	Escalated to the sponsor as a re-sequenced date with the coverage number attached; made MPN a standing metric rather than a task.	S6
8	**How would you decompose a network placement problem (e.g., inbound placement cost vs. speed)?**	Master §3.99 opening: book of record and metric → stocks and flows → broken invariant → design → sequence; then lead-time and forecast error at the lane × node grain as the two levers.	Master §3.99
9	**How do you validate a model before the business acts on it?**	Parallel run against the source (daily reconciliation to SAP; manual sample per cohort); a published gate; the old process stays alive until the match holds.	S6 (S10 pointer — Josyula's)
10	**What is your quantitative background?**	MIT Supply Chain MicroMasters + Supply Chain Network Design certificate — acquired to design what I had been implementing; the formulas are that toolkit, labeled. Never "I build models for a living."	Master §4.1 toolkit line
11	**Tell me about a time you improved a process end to end (Plan → Deliver).**	S6 as the Plan redesign (definition → gate → alert → release rule) with S8 as the Source/Deliver parameter feeding it; the two programs stay separate in dollars.	S6 + S8
12	**Another example of Dive Deep?**	Pointers only: S10 (Austria — bar raiser's) and S7 (phantom math — unallocated; no BOM-explosion content). If pressed, S7 at the level of "six sites, three waves, five months, zero auto-loaded revision overlaps."	S7 pointer

Cheat-numbers box — canon values with basis labels

Canon numbers for S6 and S8 with their basis labels, from the master packet's §4.1 ledger.

METRIC	VALUE TO SAY	BASIS	NEVER
S6 forecast accuracy	70% → 85% on scoped class-A SKUs	record	claiming I owned the forecast
S6 safety stock	$2M released under the both-errors-improved rule	record, one-time	"$2M per year"; adding to S8
S6 recurring	≈$0.8M/yr (carry 0.2 · expedites 0.3 · planner capacity 0.3); range $0.5–1.2M	modeled	presenting as record
S6 timing / scope	shortage alert live in 3 months; eight domains; six sites; adoption 8×; planner effort −50%; shipment data 60% complete at start	record	pasting "3 months" onto the six-site ingest (S7)
S6 alert rule	days-of-cover ≤ lane lead-time p90	record mechanism	—
S8 lanes / accuracy	200 lanes; 99% lead-time accuracy at the p99 promise	record	—
S8 release	580K units / ~$1.5M, zero stockouts in the watch window	record, one-time	"$1.5M to $2M"; annualizing it
S8 recurring	≈$0.6M/yr (carry 0.3 at 20% · expedites 0.3); range $0.3–1.4M	modeled	presenting as record
S8 unit value	≈$2.59/unit	derived from the record pair	—
Site count	"six sites"	record	"six NA sites"
Toolkit	MIT Supply Chain MicroMasters + Network Design certificate; formulas are illustrations	labeled	"I build supply chain models for a living"
S7 (pointer only)	six sites, three waves, 5 months; hardest site 6 weeks vs 3-week average	record	"500+ product families"; 60→95% inside S7; any BOM-explosion walk

Never-say, one glance before the call.

- Never add $2M (S6) and ~$1.5M (S8) into one number; never "$1.5M to $2M."
- Never "six NA sites" — "six sites." Never paste S6's "3 months" onto the six-site ingest.
- Never claim ownership of the demand forecast or the distributions — measurement, definitions, rule and policy were mine.
- Never present the safety-stock formula as a program artifact; MRP math at finished-goods grain only — no BOM explosion.
- Never present planner capacity as cash — time redeployed, not headcount removed.
- S10 (Austria) is the bar raiser's; S11/S12, S2/S18, S1/S3 belong to Tripathy, Goforth and Lan — pointers only here.